The Integrated Toroidal–Syntropic Model: Core Architecture, Conditional Weak-Field Limit, and Open Research Gates

Brendon Boyd
Independent Researcher, Western Australia

Version 12.0-alpha.1 — Technical Correction Pass — July 17, 2026

Research-stage status. This document reconstructs a coherent ITSM field hierarchy and records both successful and failed research gates. It is not a completed alternative to ΛCDM and does not claim that the open matching, screening, lensing, disk, perturbation, or driven-anisotropy problems have been solved.

Abstract

The Integrated Toroidal–Syntropic Model (ITSM) is reconstructed here as a layered effective framework rather than a single scalar interpolation formula. The recovered architecture contains a finite-density complex condensate, an emergent infrared phonon sector, distinct matter–plenum and reservoir–plenum exchange currents, compact three-torus boundary conditions, and explicitly open wake, circulation, shape-modulus, screening and relativistic sectors. A cubic-gradient phonon action proportional to $Y^{3/2}$ conditionally yields the desired square-root acceleration in a quasi-static high-symmetry limit; it is not derived from the analytic Born–Infeld term used in the preceding manuscript. On compact T^3 the local source must be compensated and realistic disks require a nonlinear periodic boundary-value solution. Separately, the validated CBR-001 lattice and biaxial backreaction calculations establish topology-dependent directional Casimir pressure but find that $H_t/H_p = 13/12$ is only a tunable transient in the tested free-field model, not a plateau or attractor. The present work therefore defines a conservative theory architecture, a strict claim-status vocabulary and a dependency-ordered programme for microscopic matching, screening, lensing, disks, wakes and driven anisotropic stress.

Contents

1 Scope, claim status, and reconstruction rule **2**
1.1 Conventions . . . 3

2 Core fields and layered architecture **3**
2.1 Metric, condensate, and finite-density frame . . . 3
2.2 Emergent phonon and matter . . . 4
2.3 Topology, reservoir, and non-equilibrium sectors . . . 4
2.4 Architecture versus prediction . . . 4

3 Ultraviolet–infrared hierarchy **5**
3.1 Candidate analytic condensate 5
3.2 Conditional infrared action 5
3.3 Why the preceding Born–Infeld claim is not retained 6
3.4 UVIR-001 pass criteria 6

4 Conservation and syntropic throughput **6**
4.1 Candidate local interaction 6
4.2 Reservoir throughput and anisotropic stress 7

5 Conditional low-acceleration matter coupling **7**
5.1 Inverse construction of the cubic-gradient operator 7
5.2 Compact toroidal source 8
5.3 Disk geometry and coefficient status 8

6 Toroidal topology and validated Casimir backreaction **9**
6.1 Periodic rectangular lattice 9
6.2 Biaxial backreaction 9
6.3 The Stage-3B ratio test 9

7 Wake, circulation, and non-equilibrium hypotheses **10**

8 Cosmological architecture and current limits **11**
8.1 What CBR-001 changes 12
8.2 Perturbations 12

9 Dependency-ordered closure gates **12**

10 Falsifiability without overclaiming **13**
10.1 Near-term theoretical falsifiers 13
10.2 Observational claims held downstream 13
10.3 Reproducibility rule 13

1 Scope, claim status, and reconstruction rule

The preceding ITSM manuscript combined an active-superfluid ontology, a Born–Infeld interpolation, an algebraic galactic force, an open-system source and multiple cosmological claims. Subsequent equation audits and numerical gates showed that these ingredients do not yet follow from one action. The purpose of this alpha is therefore architectural: it states what the model is, which results are presently supportable, and what must still be calculated.

Four labels are used throughout:

Derived. A checked result that follows from declared assumptions within its stated domain.

Conditional. A result that also requires a named postulate, matching hypothesis, background choice or symmetry limit.

Open. A well-defined derivation or numerical test remains incomplete.

Rejected. The stated derivation or implementation failed and is not live support for the framework.

These labels apply to derivations rather than to broad themes. For example, topology-dependent Casimir stress is derived for the declared periodic scalar calculation, while a universal 13/12 expansion ratio from cycle counting is rejected. A rejected shortcut does not reject T^3 topology; likewise, a successful local effective action does not establish a relativistic or cosmological completion.

1.1 Conventions

We use metric signature $(-,+,+,+)$ and natural units $c = \hbar = 1$ except where factors are restored explicitly, as in the Casimir lattice energy. The reduced Planck mass and gravitational coupling satisfy

$$M_{\mathrm{P}}^{-2} = 8\pi G, \qquad \kappa = 8\pi G. \tag{1}$$

These conventions make the covariant $Y^{3/2}$ coefficient and the static cubic-gradient coefficient directly comparable.

Table 1: High-level status at the start of the core recovery.

Component	Status	Boundary of the result
Finite-density plenum	Conditional	Physical postulate; microscopic action open
Compact T^3 topology	Conditional	Adopted boundary condition; scale open
$Y^{3/2}$ square-root branch	Conditional	Local, quasi-static, high symmetry
Present Born–Infeld origin	Rejected	Does not give the required IR response
$C_{\mathrm{proj}} = 2/3$ local vertex	Conditional	Geometric motivation; matching open
Rectangular-T^3 Casimir tensor	Derived	Periodic free scalar with stated moduli
Persistent 13/12 from free-scalar Casimir stress	Rejected	CBR-001 finds only transient crossings
Persistent 13/12 from a derived driven ITSM sector	Open	Requires TOP/WAK/reservoir closure and CBR-002
Screening, PPN and lensing	Open	Relativistic completion absent
Realistic disk field solution	Open	Periodic nonlinear PDE required
Driven reservoir anisotropy	Open	Constitutive conversion not derived

The reconstruction rule is that no observational claim may have a stronger status than its upstream field sector. Historical calculations remain useful as diagnostics and provenance, but they are not promoted into the core merely because their numerical values are suggestive.

2 Core fields and layered architecture

2.1 Metric, condensate, and finite-density frame

The gravitational variable is the spacetime metric $g_{\mu\nu}$. The candidate plenum order parameter is complex,

$$\Phi = \frac{\rho}{\sqrt{2}} e^{i\Theta}, \tag{2}$$

so that amplitude fluctuations, phase excitations, zeros of the order parameter and winding sectors can be represented within one microscopic language. This is important because a smooth single real potential is locally irrotational, whereas the historical ITSM ontology relies on circulation, vortices and toroidal and poloidal currents.

A unit timelike vector U^μ denotes the rest frame of the finite-density state,

$$U^\mu U_\mu = -1, \qquad h^{\mu\nu} = g^{\mu\nu} + U^\mu U^\nu. \tag{3}$$

Whether U^μ is derived from Θ, from a foliation field, or from a coarse-grained fluid description is an open choice that must be settled by the relativistic gate.

2.2 Emergent phonon and matter

The infrared phonon is denoted ψ. It is an excitation of the finite-density phase, not automatically the same variable as a fundamental vacuum scalar at all scales. Matter fields are collected in Ψ_m. The matter–phonon interaction must be derived once, then used consistently both in the phonon source and in the matter stress-energy exchange.

2.3 Topology, reservoir, and non-equilibrium sectors

Spatial slices are assigned compact T^3 boundary conditions with lengths L_i. Their ratios are shape moduli. The global topology controls allowed modes and winding; it does not directly fix every local Wilson coefficient.

An explicit reservoir sector R permits the observable matter–plenum system to be open while the complete covariant system remains conserved. A wake or memory variable W and a shape sector are permitted research sectors, not assumed completed degrees of freedom.

The schematic inventory is

$$S_{\text{complete}} = S_{\text{EH}} + S_\Phi + S_T + S_m + S_R + S_{\text{int}} + S_W, \tag{4}$$

where S_W is included only if a causal and stable wake description is found. At low energy, $S_\Phi + S_{\text{int}}$ may be matched to an infrared phonon action S_{IR}. One must not add the UV and IR descriptions as independent propagating copies in the same regime; they are Wilsonian descriptions linked by matching.

2.4 Architecture versus prediction

Equation (4) is a sector map, not a declaration that each term has been derived. The architecture recovers the original active and rotational character of ITSM while making the missing links explicit. In particular,

$$\text{energy throughput} \not\Rightarrow \text{anisotropic pressure} \tag{5}$$

without a shape-dependent conversion law, and

$$\text{compact topology} \not\Rightarrow C_{\text{proj}} = \frac{2}{3} \quad \text{or} \quad \frac{H_t}{H_p} = \frac{13}{12} \tag{6}$$

without separate matching and backreaction calculations.

3 Ultraviolet–infrared hierarchy

3.1 Candidate analytic condensate

A representative microscopic starting point is

$$S_\Phi = \int \mathrm{d}^4x\sqrt{-g}\left[-\nabla_\mu\Phi^*\nabla^\mu\Phi - m^2|\Phi|^2 - \frac{\lambda_4}{2}|\Phi|^4 - \frac{\lambda_6}{3\Lambda^2}|\Phi|^6 + \cdots\right]. \tag{7}$$

Equation (7) is a matching candidate, not the canonical ITSM action. Its branch structure, finite-density state, stability and relativistic characteristics must be evaluated before adoption.

Equation (7) does not by itself select a finite-density condensate. The background must additionally carry a fixed Noether charge or chemical potential, as in relativistic superfluid effective theory [1]. With the signature declared in Sec. 1.1, write

$$\Theta(x) = \mu t + \vartheta(x), \qquad \nabla_\mu\Theta = -\mu U_\mu + \nabla_\mu\vartheta. \tag{8}$$

For a homogeneous amplitude the phase gradient produces an effective potential

$$V_{\mathrm{eff}}(\rho;\mu) = V(\rho) - \frac{1}{2}\mu^2\rho^2, \tag{9}$$

subject to the sign convention of the final microscopic action. The phonon reduction is valid only if V_{eff} has a stable nonzero minimum ρ_0, with $V'_{\mathrm{eff}}(\rho_0) = 0$ and $V''_{\mathrm{eff}}(\rho_0) > 0$.

The sextic term is of interest because a three-body density interaction can generate the power required by the infrared gate. In a Thomas–Fermi description, consider

$$\mathcal{L}(n, X) = nX - \frac{\lambda_3}{3}n^3. \tag{10}$$

The algebraic density equation gives

$$n = \sqrt{\frac{X}{\lambda_3}}, \tag{11}$$

and substitution yields

$$P(X) = \frac{2}{3\sqrt{\lambda_3}}X^{3/2}. \tag{12}$$

This establishes a plausible mechanism for the power law, not the complete relativistic ITSM matching. Gradient corrections, sign choices, the domain of X, background stability and normalization remain open. More generally, $P(X)$ phonon actions arise as the leading derivative description of finite-density superfluids [1, 2].

3.2 Conditional infrared action

Define the dimensionless spatial invariant

$$Y = \frac{h^{\mu\nu}\nabla_\mu\psi\nabla_\nu\psi}{a_0^2}. \tag{13}$$

The conditional low-acceleration operator is

$$\mathcal{L}_{\psi,\mathrm{IR}} = -\frac{2C_{\mathrm{IR}}}{3}M_{\mathrm{P}}^2a_0^2Y^{3/2} + \mathcal{O}(Y^2, \nabla^2/\Lambda_{\mathrm{IR}}^2). \tag{14}$$

Here C_{IR} is a Wilson coefficient in a chosen field normalization. It cannot be identified with a local observable coupling until the matter vertex has been normalized in the same convention.

3.3 Why the preceding Born–Infeld claim is not retained

The preceding manuscript used an analytic kinetic expression whose derivative approaches a nonzero constant at small kinetic invariant. A sourced field equation is therefore linear at leading order in that limit. The desired square-root response instead requires a kinetic derivative proportional to the field-gradient magnitude, as supplied by Eq. (14). The Born–Infeld expression may be investigated for a separate UV or regulating role, but it is not the present derivation of the galactic branch.

3.4 UVIR-001 pass criteria

A successful match must demonstrate:

1. a stable finite-density background;
2. a heavy amplitude mode that can be integrated out;
3. the sign and branch of Eq. (14);
4. positive compressibility and Hamiltonian in the domain of use;
5. a controlled sound cone and strong-coupling scale;
6. the coefficient map to G, a_0 and the normalized matter vertex.

Until these checks pass, Eq. (14) is a conditional infrared completion rather than a derived consequence of Eq. (7).

4 Conservation and syntropic throughput

The observable matter–plenum subsystem may exchange energy and momentum with a reservoir sector external to that observable subsystem but contained within the complete conserved theory. The complete system must satisfy the contracted Bianchi identity. We therefore distinguish two currents:

$$\nabla_\mu T_m^{\mu\nu} = Q^\nu_{\rm mp}, \tag{15}$$

$$\nabla_\mu T_P^{\mu\nu} = -Q^\nu_{\rm mp} + Q^\nu_{\rm syn}, \tag{16}$$

$$\nabla_\mu T_R^{\mu\nu} = -Q^\nu_{\rm syn}. \tag{17}$$

Adding Eqs. (15)–(17) gives

$$\nabla_\mu \left(T_m^{\mu\nu} + T_P^{\mu\nu} + T_R^{\mu\nu}\right) = 0. \tag{18}$$

$Q^\nu_{\rm mp}$ is the local matter–plenum exchange generated by the same interaction that sources the phonon. $Q^\nu_{\rm syn}$ is reservoir throughput. Treating them as one undifferentiated Q^ν hides two different physical jobs and prevents a clean local/cosmological split.

4.1 Candidate local interaction

A universal conformal matter metric illustrates how a force and an exchange current can share an origin,

$$\widetilde{g}_{\mu\nu} = e^{2C_m\psi/c^2} g_{\mu\nu}, \qquad S_m = S_m[\widetilde{g}_{\mu\nu}, \Psi_m]. \tag{19}$$

For pressureless matter its nonrelativistic limit contains

$$\mathcal{L}_{\rm int}^{\rm NR} = -C_m \rho_b \psi, \tag{20}$$

and its exchange current is proportional to the matter trace times $\nabla^\nu \psi$. This is a candidate matter coupling, not a complete relativistic phenomenology: a purely conformal rescaling does not by itself provide the required lensing.

4.2 Reservoir throughput and anisotropic stress

In the plenum rest frame a general stress tensor includes

$$T_P^{\mu\nu} = \rho_P U^\mu U^\nu + p_P h^{\mu\nu} + \pi^{\mu\nu}. \tag{21}$$

The traceless spatial tensor $\pi^{\mu\nu}$, not the energy-transfer vector alone, carries directional stress. In a biaxial state one may define

$$\Pi = p_t - p_p, \qquad \pi^i{}_j = \mathrm{diag}\left(-\frac{2\Pi}{3}, \frac{\Pi}{3}, \frac{\Pi}{3}\right). \tag{22}$$

For the biaxial expansion used below, the shear tensor is

$$\sigma^i{}_j = \mathrm{diag}\left(-\frac{2\delta}{3}, \frac{\delta}{3}, \frac{\delta}{3}\right), \qquad \pi^{\mu\nu}\sigma_{\mu\nu} = \frac{2}{3}\Pi\delta. \tag{23}$$

The last equality is the exact anisotropic work term for these conventions. A relation such as $Q^\nu_{\rm syn} \propto \Theta U^\nu$ injects energy along the background flow but does not determine Π. A geometry-dependent action or constitutive law is required to convert throughput into shear.

No fixed redshift law for $Q^\nu_{\rm syn}$ is adopted in this alpha. Its background and perturbation dependence must follow from the reservoir model, and local decoupling must be tested rather than asserted from $\Theta \to 0$.

5 Conditional low-acceleration matter coupling

5.1 Inverse construction of the cubic-gradient operator

Nonlinear Poisson equations derived from nonquadratic potential actions have a standard precedent in AQUAL-type theories [3]. Here we isolate the coefficient normalization required by the ITSM infrared branch. Let $q = |\nabla\psi|$ and consider the static phonon action

$$S_\psi = \int \mathrm{d}t\, \mathrm{d}^3x \left[-K(q) - C_m \rho_b \psi\right]. \tag{24}$$

Variation yields

$$\nabla \cdot \left[\frac{K'(q)}{q} \nabla\psi\right] = C_m \rho_b. \tag{25}$$

The required cubic-gradient operator is parameterized as

$$K(q) = \frac{C_{\rm IR}}{12\pi G a_0} q^3, \qquad C_{\rm IR} > 0. \tag{26}$$

The complete static weak-field action in this normalization is

$$S_{\mathrm{WF}} = \int \mathrm{d}t\, \mathrm{d}^3x \Bigg[-\frac{|\nabla\Phi_N|^2}{8\pi G} - \rho_b\Phi_N - \frac{C_{\mathrm{IR}}}{12\pi G a_0}|\nabla\psi|^3 - C_m\rho_b\psi \Bigg]. \tag{27}$$

For independent positive coefficients C_m and C_{IR}, the sourced equation is

$$\nabla\cdot\left[\frac{|\nabla\psi|}{a_0}\nabla\psi\right] = 4\pi G\frac{C_m}{C_{\mathrm{IR}}}\rho_b. \tag{28}$$

For spherical symmetry this gives

$$|\nabla\psi|^2 = \frac{C_m}{C_{\mathrm{IR}}}a_0 g_N. \tag{29}$$

Since the physical acceleration is $g_P = C_m|\nabla\psi|$, the invariant observable square-root coefficient is

$$g_P = C_{\mathrm{obs}}\sqrt{a_0 g_N}, \qquad \boxed{C_{\mathrm{obs}} = \frac{C_m^{3/2}}{\sqrt{C_{\mathrm{IR}}}}}. \tag{30}$$

Only under the normalization choice $C_m = C_{\mathrm{IR}} = C$ does $C_{\mathrm{obs}} = C$. That convenient choice is not a physical derivation of $C = 2/3$; MAT-001 must calculate the invariant combination in Eq. (30).

5.2 Compact toroidal source

On compact T^3 there is no boundary at spatial infinity, and a periodic divergence integrates to zero. Write

$$\rho_b = \bar\rho_b + \delta\rho_b, \qquad \psi = \bar\psi(t) + \varphi(t,\mathbf{x}), \tag{31}$$

with zero spatial averages for $\delta\rho_b$ and φ. The local periodic equation has the form

$$D_i\left[\frac{|D\varphi|}{a_0}D^i\varphi\right] = 4\pi G\frac{C_m}{C_{\mathrm{IR}}}\delta\rho_b + \mathcal{S}_Q, \qquad \int_{T^3}\left(4\pi G\frac{C_m}{C_{\mathrm{IR}}}\delta\rho_b + \mathcal{S}_Q\right)\mathrm{d}V = 0. \tag{32}$$

Here $\mathcal{S}_Q \equiv \mathcal{S}_Q[\delta Q_{\mathrm{syn}}]$ is a conditional zero-mean scalar source representing the inhomogeneous projection of reservoir–plenum exchange into the phonon equation. Its functional map from the vector exchange current has not been derived; setting $\mathcal{S}_Q = 0$ is the local-static baseline rather than a claim that reservoir perturbations vanish identically. The homogeneous mode belongs to the cosmological/open-system problem. A localized spherical solution is therefore the small-region, compensated-overdensity limit of Eq. (32), not an isolated global cosmology.

5.3 Disk geometry and coefficient status

For a general density distribution, equality of divergences permits

$$\frac{|\nabla\varphi|}{a_0}\nabla\varphi = \frac{C_m}{C_{\mathrm{IR}}}\nabla\Phi_N + \nabla\times\mathbf{H}. \tag{33}$$

The solenoidal field vanishes in sufficient symmetry but is generally nonzero for a finite disk. Consequently a pointwise square-root prescription is a historical algebraic approximation. DISK-001 must solve Eq. (32) for realistic baryonic sources before $C_{\mathrm{obs}} = 2/3$ and $C_{\mathrm{obs}} \simeq 1$ can be fairly compared.

The separate coefficients change under a rescaling of ψ, while the physical response is carried by the invariant combination $C_{\mathrm{obs}} = C_m^{3/2}/\sqrt{C_{\mathrm{IR}}}$. The trace ratio 2/3 remains a geometric motivation and a matching hypothesis, not a completed local Wilson coefficient.

6 Toroidal topology and validated Casimir backreaction

6.1 Periodic rectangular lattice

For a massless scalar with periodic lengths (L_1, L_2, L_3), the renormalized energy density used in CBR-001 is the rectangular-periodic Epstein-lattice result developed in Ref. [4]:

$$\rho_{\mathrm{Cas}} = -\frac{\hbar c}{2\pi^2} \sum_{\mathbf{n}\in\mathbb{Z}^3\setminus\{0\}} \frac{1}{[(n_1L_1)^2 + (n_2L_2)^2 + (n_3L_3)^2]^2}. \tag{34}$$

The directional pressures follow thermodynamically,

$$p_i = -\frac{1}{L_jL_k}\frac{\partial E_{\mathrm{Cas}}}{\partial L_i}, \qquad E_{\mathrm{Cas}} = L_1L_2L_3\rho_{\mathrm{Cas}}. \tag{35}$$

The Stage-1 and Stage-2 calculations validate cutoff convergence, cubic isotropy, pressure derivatives, the massless trace relation, dimensional scaling and the sign change of $p_t - p_p$ across the cubic shape. Thus compact shape-dependent anisotropic stress is a concrete mechanism. It is not a universal cycle-counting coefficient. A recent direct semiclassical calculation likewise derives a finite topology-dependent stress tensor on T^3 and its anisotropic de Sitter backreaction [7].

6.2 Biaxial backreaction

For

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a_p^2\mathrm{d}x^2 + a_t^2(\mathrm{d}y^2 + \mathrm{d}z^2), \tag{36}$$

define

$$H = \frac{H_p + 2H_t}{3}, \quad \delta = H_t - H_p, \quad r = \frac{a_t}{a_p}. \tag{37}$$

Then

$$\dot{\delta} + 3H\delta = \kappa(p_t - p_p), \qquad \dot{r} = r\delta, \tag{38}$$

with the Hamiltonian constraint

$$3H^2 - \frac{\delta^2}{3} = \kappa\rho_{\mathrm{total}}. \tag{39}$$

These are the biaxial specialization of the standard Bianchi-I system; see, for example, Ref. [5]. The Stage-3A solver reproduces the perturbative solution and preserves the constraint and Casimir continuity equation to substantially better than the specified tolerances.

6.3 The Stage-3B ratio test

Let

$$x = \frac{\delta}{H}, \qquad q = \frac{H_t}{H_p} = \frac{1 + x/3}{1 - 2x/3}. \tag{40}$$

The historical target $q = 13/12$ corresponds to

$$x_\star = \frac{3}{38}. \tag{41}$$

The small-source fixed-background solution is

$$x(N) = S\left(e^{-3N} - e^{-4N}\right), \tag{42}$$

with maximum $(27/256)S$ at $N = \ln(4/3)$. Merely reaching $x_\star$ therefore requires

$$S \geq \frac{128}{171} \simeq 0.748538, \tag{43}$$

already outside the strict small-perturbation regime.

The nonlinear root search finds target-reaching amplitudes for five of the tested initial shapes. Every case is classified as a transient crossing, with only approximately 0.27–0.30 e-folds within one percent of the target. Four thresholds are nonperturbative and one is marginal by the stated Casimir-fraction diagnostic. Every valid candidate returns to $q = 1$ by $N = 20$; no quasi-plateau or attractor is found.

There is also a kinematic warning. Since

$$\frac{\mathrm{d} \ln r}{\mathrm{d}N} = x, \tag{44}$$

a constant $q \neq 1$ implies continuous evolution of the shape ratio. It cannot be a fixed torus shape. A persistent anisotropic state would have to be a driven solution, not a static geometric ratio.

For a desired trajectory $x(N)$, the required anisotropic stress is

$$\Pi_{\text{req}} = \frac{H^2}{\kappa} \left[\frac{\mathrm{d}x}{\mathrm{d}N} + x \left(3 + \frac{\mathrm{d} \ln H}{\mathrm{d}N} \right) \right]. \tag{45}$$

In de Sitter with constant $x = 3/38$ this becomes

$$\Pi_{\text{req}} = \frac{9H^2}{38\kappa}. \tag{46}$$

For the CBR-001 physical lengths $L_p = L_\star a r^{-2/3}$ and $L_t = L_\star a r^{1/3}$, the anisotropic Casimir source is more precisely

$$\Pi_{\text{Cas}}(a, r) = \frac{\hbar c}{L_\star^4} a^{-4} r^{8/3} \left[\widehat{p}_t(r) - \widehat{p}_p(r) \right] \equiv \frac{\hbar c}{L_\star^4} a^{-4} F_\Pi(r). \tag{47}$$

Thus the explicit scale dependence is radiation-like, while the shape function evolves whenever r evolves. In every validated free-field trajectory the source nevertheless becomes negligible and cannot maintain Eq. (45). A future CBR-002 test must derive any additional shape, driven or memory stress without inserting the target ratio into a coefficient.

7 Wake, circulation, and non-equilibrium hypotheses

The original ITSM described baryonic matter as exciting a torsional or acoustic wake in a rotational medium. The static equation Eq. (32) has no independent memory: for specified boundary data it tracks the instantaneous source. A persistent, advected or detached wake therefore requires additional dynamics.

A schematic relaxation equation is

$$\tau_W U^\mu \nabla_\mu W + W = \mathcal{F}[\psi, \rho_b, r, Q_{\text{syn}}], \tag{48}$$

or, if wave propagation is essential, a hyperbolic field equation with a second time derivative. Equation (48) is not adopted as a law. It identifies the quantities that WAK-001 must derive or bound. Its first-order form is motivated only as a causal-relaxation template of the Israel–Stewart type [6]; it is not a derivation of an ITSM constitutive law.

A viable wake sector must satisfy all of the following:

1. a well-posed initial-value problem;
2. causal characteristic speeds in the physical frame;
3. a positive or bounded energy accounting;
4. a controlled static limit reproducing the IR field equation;
5. explicit momentum exchange with matter and the reservoir;
6. decay or transport laws rather than indefinite unsupported memory.

Similarly, circulation is represented naturally by the phase in Eq. (2),

$$\oint \nabla_i \Theta \, \mathrm{d}x^i = 2\pi n, \qquad n \in \mathbb{Z}, \tag{49}$$

where zeros of ρ permit defect cores. The relationship between these microscopic winding sectors and a global toroidal current is the VOR-001 gate. No black-hole, lensing or galactic coefficient follows from the winding number without further dynamics.

For driven anisotropy one may organize the bookkeeping as

$$\Pi_{\mathrm{total}} = \Pi_{\mathrm{Cas}} + \Pi_{\mathrm{shape}} + \Pi_{\mathrm{driven}} + \Pi_{\mathrm{memory}}. \tag{50}$$

Only Π_{Cas} has been calculated in the present programme. The other terms are labels for research sectors. Writing Eq. (50) does not establish that they exist or have the required sign and magnitude.

8 Cosmological architecture and current limits

The homogeneous condensate, reservoir and topology moduli must define one fiducial cosmological background before CMB, growth, age or distance results are calculated. Earlier analyses used several dataset-conditioned values of the syntropic exponent or effective equation of state and sometimes mixed them across diagnostics. Those outputs remain provenance and cannot be treated as one canonical cosmology.

At background level, a conservative sector decomposition has the form

$$\dot{\rho}_P + 3H(\rho_P + p_P) + \pi^{\mu\nu}\sigma_{\mu\nu} = Q, \tag{51}$$

$$\dot{\rho}_R + 3H(\rho_R + p_R) = -Q, \tag{52}$$

For the biaxial conventions of Sec. 6, Eq. (23) gives the exact specialization

$$\dot{\rho}_P + 3H(\rho_P + p_P) + \frac{2}{3}\Pi\delta = Q. \tag{53}$$

The total continuity equation follows by summing the sectors.

No predetermined law such as $Q \propto (1+z)^{-3}$ is adopted here. In particular, $(1+z)^{-3} = a^3$ grows toward late time and cannot simultaneously be described as an early-universe enhancement. COS-001 must derive the source and fit one background with an explicit parameter inventory.

8.1 What CBR-001 changes

CBR-001 does not remove topology-dependent cosmological stress. It shows that the tested passive free-field source has the evolving form $\Pi_{\rm Cas} = a^{-4}F_{\Pi}(r)$ and becomes unable to support a persistent 13/12 directional expansion ratio. Therefore

- $H_0 = 72.97\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is not a parameter-free ITSM prediction;
- assigning Planck and SH0ES to opposite directional extrema is not a derived observation model;
- any future directional expansion must be passed through actual survey window functions;
- a driven anisotropic state requires a derived source beyond free Casimir stress.

8.2 Perturbations

PERT-001 must vary the same total action and exchange currents used for the background. Effective CAMB or CLASS mappings are useful diagnostics, but they do not replace the derivation of scalar, vector and tensor perturbations, initial conditions, stability and observable transfer functions. Until that gate closes, CMB-peak and S_8 values are diagnostic calculations rather than canonical predictions.

9 Dependency-ordered closure gates

Table 2: Research gates required before restoration of strong predictions.

Gate	Question	Pass condition
UVIR-001	Does a stable condensate yield $Y^{3/2}$?	Controlled amplitude integration and coefficient match
MAT-001	What is the normalized matter vertex?	Force and exchange from one action; C_m calculated
SCR-001	Is the force screened at high gradient?	Ephemeris and binary constraints satisfied
LEN-001	What metric do matter and light see?	PPN, lensing and wave propagation derived
DISK-001	What is the exact disk response?	Validated periodic nonlinear solver and curl field
TOP-001	How do shape moduli evolve?	Covariant moduli action and stable dynamics
VOR-001	How is global circulation carried?	Winding/defect sector with finite energy
WAK-001	Can wakes retain causal memory?	Hyperbolic or relaxation model with energy accounting
CBR-002	Can derived sectors sustain anisotropy?	Untuned non-equilibrium solution and stability test
COS-001	What is the fiducial background?	One coherent fit from declared equations
PERT-001	What are the perturbation equations?	Action-derived stable transfer system

The dependency order is

$$\mathrm{UVIR} \to \mathrm{MAT} \to (\mathrm{SCR}, \mathrm{LEN}) \to \mathrm{DISK} \to (\mathrm{TOP}, \mathrm{VOR}, \mathrm{WAK}, R) \to \text{CBR-002} \to (\mathrm{COS}, \mathrm{PERT}). \tag{54}$$

This ordering is deliberately restrictive. For example, running CBR-002 with an arbitrary source adjusted to hold 13/12 would test a tuning procedure, not ITSM. Likewise, fitting the algebraic disk law more aggressively would not answer whether the actual periodic field equation changes the inferred coupling.

10 Falsifiability without overclaiming

The recovery replaces spectacular but weakly derived forecasts with a sequence of gate-level falsifiers.

10.1 Near-term theoretical falsifiers

1. If no stable finite-density branch can yield the required phonon power and sign, UVIR-001 rejects the proposed condensate origin of the local law.
2. If a normalized matter vertex cannot satisfy galactic and local bounds within one action, MAT-001/SCR-001 reject the universal coupling architecture.
3. If no relativistic completion produces acceptable PPN and lensing potentials, the local acceleration law cannot serve as a gravitational model.
4. If the periodic disk solution preserves the empirical tension of the algebraic $C_{\mathrm{obs}} = 2/3$ implementation, disk geometry cannot rescue that matching hypothesis.
5. If derived reservoir, shape and wake sectors have no stable driven anisotropic solution, CBR-002 rejects persistent topological Hubble anisotropy within that architecture.

10.2 Observational claims held downstream

NANOGrav spectral features require a derived eigenmode spectrum and detector response. Cluster offsets require causal wake dynamics, a collision calculation and a lensing metric. Stellar-population claims require an independent star-formation model. CMB and growth require a coherent background and perturbation theory. None is presented here as a clean framework-level falsifier.

10.3 Reproducibility rule

Every future numerical claim must identify its source equations, code entry point, data version, parameter inventory, validation controls and a machine-readable summary. A reported statistic must be explicitly computed by the cited pipeline. Negative gate results, including the free-field 13/12 result, remain part of the model record rather than being hidden by a new ansatz.

Conclusion

The recovery preserves the defining ITSM idea—a driven, finite-density, rotational medium with compact topology and sector exchange—while removing the unsupported shortcuts that had come to carry that idea. The local $Y^{3/2}$ weak-field branch and the rectangular-T^3 Casimir tensor are concrete advances. Their microscopic connection to a condensate, their relativistic completion and their consequences for realistic systems remain research gates. Future predictive claims must be downstream of those gates.

References

[1] D. T. Son, "Low-energy quantum effective action for relativistic superfluids," arXiv:hep-ph/0204199 (2002), https://arxiv.org/abs/hep-ph/0204199.

[2] D. T. Son and M. Wingate, "General coordinate invariance and conformal invariance in nonrelativistic physics: Unitary Fermi gas," *Annals of Physics* **321**, 197–224 (2006), doi:10.1016/j.aop.2005.11.001, https://arxiv.org/abs/cond-mat/0509786.

[3] J. Bekenstein and M. Milgrom, "Does the missing mass problem signal the breakdown of Newtonian gravity?," *Astrophysical Journal* **286**, 7–14 (1984), doi:10.1086/162570.

[4] A. Edery, "Multidimensional cut-off technique, odd-dimensional Epstein zeta functions and Casimir energy of massless scalar fields," *Journal of Physics A* **39**, 685–712 (2006), doi:10.1088/0305-4470/39/3/017, https://arxiv.org/abs/math-ph/0510056.

[5] G. F. R. Ellis and M. A. H. MacCallum, "A class of homogeneous cosmological models," *Communications in Mathematical Physics* **12**, 108–141 (1969), doi:10.1007/BF01645908.

[6] W. Israel and J. M. Stewart, "Transient relativistic thermodynamics and kinetic theory," *Annals of Physics* **118**, 341–372 (1979), doi:10.1016/0003-4916(79)90130-1.

[7] A. Negro *et al.* (COMPACT Collaboration), "Quantum signatures of cosmic topology: How Casimir backreaction transmits isotropy violation," arXiv:2603.12319 (2026), https://arxiv.org/abs/2603.12319.